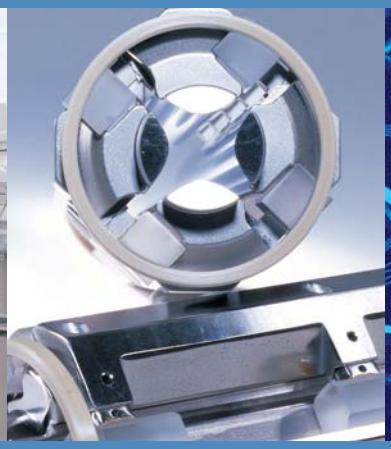
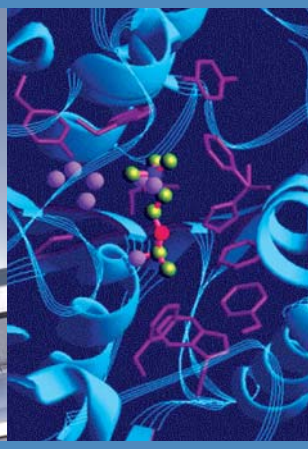
# Finnigan™ Kiel IV Carbonate Device

## Preinstallation Requirements Guide

Revision A - 119 6850







Analyze • Detect • Measure • Control™

Thermo ELECTRON CORPORATION

# Finnigan™ Kiel IV Carbonate Device

## Preinstallation Requirements Guide

Revision A - 119 6850

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Declaration of conformity according to EN 45014
Dichiarazione di conformità alla EN 45014

**Name des Herstellers:**
manufacturers name
nome produttore

Thermo Electron (Bremen) GmbH

**Adresse des Herstellers:**
manufacturers address
indirizzo produttore

Hanna-Kunath-Strasse 11
28199 Bremen
Germany

**erklärt, dass das Produkt**
declares that the following product
dichiara che il seguente prodotto

## KIEL IV

**Carbonate Device**

**mit den folgenden Produktspezifikationen übereinstimmt:**
complies with the following product specifications
rispetta le seguenti specifiche del prodotto

| | |
|---|---|
| **EMV (Störemissionen):**<br>EMC (emissions)<br>EMC (emissioni) | EN 50081-1; EN 55022 class B |
| **EMV (Störfestigkeit):**<br>EMC (immunity)<br>EMC (immunità) | EN 61000-3-2, -3; EN 61000-4-2, -3, -4, -5, -6, -11; EN 61000-6-2; EN 50204 |
| **Elektrische Sicherheit:**<br>electrical safety<br>sicurezza elettrica | EN 61010-1 |
| **Ergänzende Informationen:**<br>complementary information<br>informazioni complementari | _____ |

Dieses Produkt erfüllt die EMV-Richtlinie 89/336/EWG und Niederspannungsrichtlinie 73/23/EWG.
This product complies with EMC directive 89/336/EEC and Low Voltage Directive 73/23/EEC.
Questo produtto rispetta la direttiva 89/336/EEC e la direttiva 73/23/EEC.

**Bremen, Germany, 23. März 2005**

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**Technischer Leiter:**
Operations Manager
Direttore fabrizione

ppa. H. Schröder

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# Read This First

Welcome to the Thermo Electron, Finnigan Kiel IV Carbonate Device system!

This *Finnigan Kiel IV Carbonate Device Preinstallation Requirements Guide* provides you with information that will assist you in planning for and preparing your lab site prior to delivery and installation of your system.

It summarizes requirements related to site, power, the reference sample and the various gases in use before operating your Finnigan Kiel IV Carbonate Device.

## Changes to the Manual

To suggest changes to this manual, please send your comments to:

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You are encouraged to report errors or omissions in the text or index. Thank you.

## Typographical Conventions

Typographical conventions have been established for Thermo Electron manuals for the following:

- Data input
- Admonitions
- Topic headings

## Data Input

Throughout this manual, the following conventions indicate data input and output via the computer:

- Messages displayed on the screen are represented by capitalizing the initial letter of each word and by italicizing each word.
- Input that you enter by keyboard is identified by quotation marks: single quotes for single characters, double quotes for strings.
- For brevity, expressions such as "choose File > **Directories**" are used rather than "pull down the File menu and choose Directories."
- Any command enclosed in angle brackets < > represents a single keystroke. For example, "press **<F1>**" means press the key labeled *F1*.
- Any command that requires pressing two or more keys simultaneously is shown with a plus sign connecting the keys. For example, "press **<Shift> + <F1>**" means press and hold the <Shift> key and then press the <F1> key.
- Any button that you click on the screen is represented in bold face letters. For example, "click on **Close**".

## Admonitions

Admonitions contain information that is important, but not part of the main flow of text.

Admonitions can be of the following types:

- **Note** – information that can affect the quality of your data. In addition, notes often contain information that you might need if you are having trouble.
- **Caution** – information necessary to protect your instrument from damage.
- **Warning** – hazards to human beings. Each Warning is accompanied by a Warning symbol.

## Topic Headings

The following headings are used to show the organization of topics within a chapter:

# Chapter Name

The following headings appear in the left column of each page:

## Second Level Topics

### Third Level Topics

#### Fourth Level Topics

## Safety and EMC Information

In accordance with our commitment to customer service and safety, these instruments have satisfied the requirements for the European CE Mark including the Low Voltage Directive.

Designed, processor and tested in an ISO9001 registered facility, this instrument has been shipped to you from our manufacturing facility in a safe condition.

**Caution** This instrument must be used as described in this manual. Any use of this instrument in a manner other than described here may result in instrument damage and/or operator injury. ▲

## Identifying Safety Information

The *Finnigan Kiel IV Carbonate Device Preinstallation Requirements Guide* contains precautionary statements that can prevent personal injury, instrument damage, and loss of data if properly followed. Warning symbols which alert the user to check for hazardous conditions. These appear throughout the manual, where applicable, and are defined in Table i on page v.

Table i. Warning Symbols

| Symbol | Description |
|---|---|
|  | **General** This general symbol indicates that a hazard is present, which if not avoided, could result in injuries. The source of danger is described in the accompanying text. ▲ |
|  | **Cold Burns Hazard** Wear protective clothing. ▲ |
|  | **Electric Shock** High voltages capable of causing personal injury are used in the instrument. The instrument must be shut down and disconnected from line power before service or repair work is performed. ▲ |
|  | **Magnetic Field** Keep away from heart pacemakers, computers, credit cards, and any other magnetically sensitive device. ▲ |
|  | **Noxious** This symbol alerts to hazards resulting from noxious fumes. ▲ |
|  | **Hot Surface / Heat** Allow heated components to cool down before servicing them! ▲ |
|  | **Poisonous Gases** This symbols points to possible danger because of poisonous gases and vapors. ▲ |

## Instrument-Specific Hazards

Every instrument has specific hazards, so be sure to read and comply with the following precautions. They will help ensure the safe, long-term use of your system.

1. Before plugging in any of the instrument modules or turning on the power, always make sure that the voltage and fuses are set appropriately for your local line voltage.

2. Only use fuses of the type and current rating specified. Do not use repaired fuses and do not short-circuit the fuse holder.

3. The supplied power cord must be inserted into a power outlet with a protective earth contact (ground). When using an extension cord, make sure that the cord also has an earth contact.

4. Do not change the external or internal grounding connections. Tampering with or disconnecting these connections could endanger you and/or damage the system.

**Caution** The instrument is properly grounded in accordance with regulations when shipped. You don't need to make any changes to the electrical connections or to the instrument's chassis to ensure safe operation. ▲

5. Never run the system without the housing on. Permanent damage can occur.

6. Do not turn the instrument on if you suspect that it has incurred any kind of electrical damage. Instead, disconnect the power cord and contact a Service Representative for a product evaluation. Do not attempt to use the instrument until it has been evaluated. (Electrical damage may have occurred if the system shows visible signs of damage, or has been transported under severe stress.)

7. Damage can also result if the instrument is stored for prolonged periods under unfavorable conditions (e.g. subjected to heat, water, etc.).

8. Always disconnect the power cord before attempting any type of maintenance.

9. Capacitors inside the instrument may still be charged even if the instrument is turned off. The superconducting magnet is still charged even if the instrument is turned off.

10. Never try to repair or replace any component of the system that is not described in this manual without the assistance of your service representative.

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# Preinstallation Requirements

This chapter contains the following topics:


- "Site Requirements" on page 2
- "Power Requirements" on page 2
- "Gas Requirements" on page 2
- "Further Requirements" on page 3
- "Reference Sample" on page 4

## Site Requirements

The Finnigan Kiel IV Carbonate Device is attached to Thermo Electron isotope ratio mass spectrometers, e.g. a Finnigan DELTA V Plus or a Finnigan DELTA V Advantage, equipped with a Dual Inlet system.

It is placed stand-alone but must be arranged next to the IRMS within a maximum distance of 30 cm. The distance to walls may not be less than 60 cm. The space required is 900 mm width x 900 mm depth. Its height is 1900 mm, and it weighs approximately 100 kg (220 lb). See Figure 1.

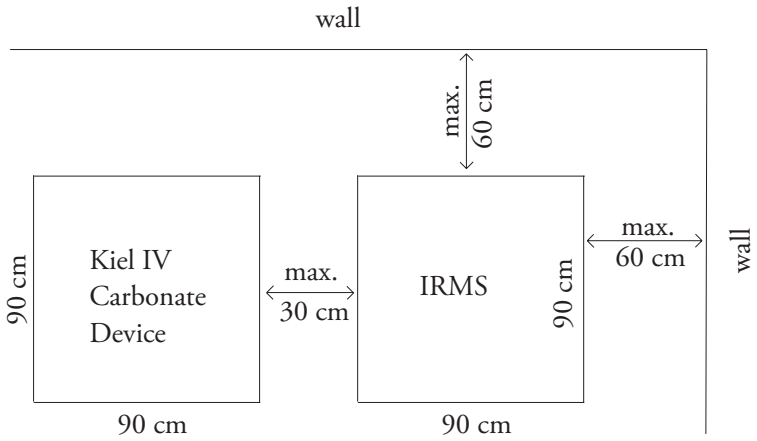



**Figure 1.** Kiel IV Carbonate Device - Site Requirements

## Power Requirements

The Finnigan Kiel IV Carbonate Device will be supplied by the IRMS line distributor. The power consumption of the IRMS will increase by 1.2 kW.

**Note** It is absolutely necessary to run your instrument without disruptions of power supply! Thus, if your local area is susceptible to corrupted power or power disruptions, an uninterruptible power supply (UPS) should be installed in your laboratory. ▲

## Gas Requirements

For operation, the Kiel IV Carbonate Device needs the gases summarized in Table 1.

**Table 1.** Kiel IV Carbonate Device - Gas Requirements

| Gas | Comment |
|---|---|
| He, $N_2$ | 1 bar as sample vial vent gas |
| liquid $N_2$ | For the liquid nitrogen-cooled trap provide approximately 0.5 L of liquid nitrogen per sample. |
| $CO_2$ | Used as reference gas. |

**Table 1.** Kiel IV Carbonate Device - Gas Requirements, continued

| Gas | Comment |
|---|---|
| Ar | Sometimes, it may be necessary to check the unit for leaks. Therefore, use an argon tank. |
| compressed air | Supplied by the compressed air distributor of the IRMS. Should be in the range between 2.8 bar and 6 bar (that is, between 40 and 87 psi). |



**Warning** All gas lines should be oil-free and preferably flame-dried. The gas lines or gas tanks should be at a distance of 1 m to 1.5 m to the instrument. ▲



**Warning** All regulators should be oil- and grease-free and be specified for gases of high purity. The supply lines should terminate with 1/8“ male Swagelok® type connectors.
Thermo Electron recommends using regulators with an outlet pressure range between 0 and 5 bar (that is, 0 and 73 psi). ▲



**Warning** All compressed air tubing and the air compressor should contain a water and oil trap. Water and oil may fill the compressed air supply and destroy the pneumatic valves! ▲

## Further Requirements

Additionally, the following items are required for the Kiel IV Carbonate Device:

- A laboratory to wash and prepare sample vials
- A fume hood, hot plate and stirrer to prepare phosphoric acid[1]
- Contaminant-free weighing instruments[1]
- Phosphorous pentoxide (500 g)[1]
- Acetone p.a. for cleaning vials and weighing instruments[1]
- A suitable micro balance to verify the installation weight specifications[1]
- A liquid nitrogen tank (approximately 30 L per day)[1]

[1]Keep available within installation distance.

# Reference Sample

To demonstrate the specified precision for small samples, it is required that the customer buys and supplies a sample of NBS-19 from the International Atomic Energy Agency or the National Bureau of Standards. Refer to Figure 2.

| Name | Nature | | Isotopic ratio | δ‰ | Reference standard |
|---|---|---|---|---|---|
| V-SMOW | water | 2H/1H | (155.761 ± 0.05) x 10e-6 (1)<br>(155.751 ± 0.08) x 10e-6 (2)<br>(155.601 ± 0.12) x 10e-6 (3) | 0 | V-SMOW |
| | | 18O/16O | (2005.20 ± 0.45) x 10e-6 (4) | 0 | V-SMOW |
| | | I70/160 | (379.91 ± 0.8) x 10e-6 (5) | 0 | V-SMOW |
| SLAP | water | 2H/1H | (89.021 ± 0.05) x 10e-6 (1)<br>(89.12 ± 0,07) x 10e-6 (2)<br>(88.88 ± 0.18) x 10e-6 (3) | -428.0 (6) | V-SMOW |
| | | 18O/16O | (1893.91 ± 0.45) x 10e-6 (7) | -55.50 (6) | V-SMOW |
| NBS-19 | calcite | 13C/12C | | 1.95 (8) | V-PDB |
| | | 180/160 | | -2.20 (8) | V-PDB |
| | | | | 28.6 (9) | V-SMOW |
| | | | **intercomparison materials** | | |
| GISP | water | 2H | | -189.73 ± 0.87 | V-SMOW |
| | | 18O | | -24.784 ± 0.075 | V-SMOW |
| NBS-18 | calcite | 13C | | -5.029 ± 0.049 | V-PDB |
| | | 18O | | -23.035 ± 0.172 | V-PDB |
| IAEA-CO-1 | calcite | 13C | | 2.48 ± 0.025 | V-PDB |
| | | 18O | | -2.437 ± 0.073 | V-PDB |
| IAEA-CO-8 | calcite | 13C | | -5.749 ± 0.063 | V-PDB |
| | | 18O | | -22.667 ± 0.187 | V-PDB |
| IAEA-CO-9 | BaCO3 | 13C | | -47.119 ± 0.149 | V-PDB |
| | | 18O | | -15.282 ± 0.093 | V-PDB |

**Figure 2.** IAEA Primary Standards

Calibrating vs. international standards requires users to have their own specimens of Primary Standards. Primary Standards are exclusively distributed by the IAEA via agencies in Europe and the US. The reference list shown in Figure 2 is taken from IAEA TECDOC 825, and the IAEA Analytical Quality Control Services Reference Materials Catalogue 2002–2003. It is very difficult to obtain homogeneity in both isotopes, $^{13}C$ and $^{18}O$.

**Note** Refer to IAEA-TECDOC-825: Reference and intercomparison materials for stable isotopes of light elements. Proceedings of a consultants meeting held in Vienna, December 1–3, 1993. International Atomic Energy Agency (IAEA). ▲

**Note** Also refer to Chapter 5.2 Environmental Level, pp. 55 in: IAEA Analytical Quality Control Services Reference Materials Catalogue 2002–2003. First edition, January 2002. Edited by Analytical Quality Control Services, International Atomic Energy Agency, P.O. Box 100, A-1400 Vienna. ▲

# Glossary

**A** ampere

**ac** alternating current

**ADC** analog-to-digital converter

**AP** acquisition processor

**APCI** atmospheric pressure chemical ionization

**API** atmospheric pressure ionization

**ASCII** American Standard Code for Information Interchange

**b** bit

**B** byte (8 b)

**baud rate** data transmission speed in events per second

**°C** degrees Celsius

**cfm** cubic feet per minute

**CI** chemical ionization

**CIP** carriage and insurance paid to

**cm** centimeter

**$cm^3$** cubic centimeter

**CPU** central processing unit (of a computer)

**CRC** cyclic redundancy check

**CRM** consecutive reaction monitoring

**<Ctrl>** control key on the terminal keyboard

**d** depth

**Da** dalton

**DAC** digital-to-analog converter

**dc** direct current

**DDS** direct digital synthesizer

**DEP™** direct exposure probe

**DS** data system

**DSP** digital signal processor

**EI** electron ionization

**EMBL** European Molecular Biology Laboratory

**<Enter>** Enter key on the terminal keyboard

**ESD** electrostatic discharge

**ESI** electrospray ionization

**eV** electron volt

**f** femto ($10^{-15}$)

**°F** degrees Fahrenheit

***.fasta* file** extension of a SEQUEST search database file

**FOB** free on board

**ft** foot

**FTP** file transfer protocol

**g** gram

**G** Gauss; giga ($10^9$)

**GC** gas chromatograph; gas chromatography

**GC/MS** gas chromatograph / mass spectrometer

**GND** electrical ground

**GPIB** general-purpose interface bus

**GUI** graphical user interface

**h** hour; height

**HPLC** high-performance liquid chromatograph

**HV** high voltage

**Hz** hertz (cycles per second)

**IAEA** International Atomic Energy Agency

**ICIS™** Interactive Chemical Information System

**ICL™** Instrument Control Language™

**ICP** inductively coupled plasma

**ICP-OES** inductively coupled plasma optical emission spectroscopy

**ID** inside diameter

**IEC** International Electrotechnical Commission

**IEEE** Institute of Electrical and Electronics Engineers

**in** inch

**I/O** input/output

**IRMS** isotope ratio mass spectrometer

**k** kilo ($10^3$, 1000)

**K** kilo ($2^{10}$, 1024)

**KEGG** Kyoto Encyclopedia of Genes and Genomes

**kg** kilogram

**l** length

**L** liter

**LAN** local area network

**lb** pound

**LC** liquid chromatograph; liquid chromatography

**LC/MS** liquid chromatograph / mass spectrometer

**LED** light-emitting diode

**LHe** liquid helium

**LN2** liquid nitrogen

**μ** micro ($10^{-6}$)

**m** meter

**m** milli ($10^{-3}$)

**M** mega ($10^6$)

**$M^+$** molecular ion

**MB** Megabyte (1 048 576 bytes)

**$MH^+$** protonated molecular ion

**min** minute

**mL** milliliter

**mm** millimeter

**MS** mass spectrometer; mass spectrometry

**MS** $MS^n$ power: where n = 1

**MS/MS** $MS^n$ power: where n = 2

**$MS^n$** $MS^n$ power: where n = 1 through 10

**m/z** mass-to-charge ratio

**n** nano ($10^{-9}$)

**NBS** National Bureau of Standards

**NCBI** National Center for Biotechnology Information (USA)

**NIST** National Institute of Standards and Technology (USA)

**OD** outside diameter

**Ω** ohm

**p** pico ($10^{-12}$)

**Pa** pascal

**PCB** printed circuit board

**PID** proportional / integral / differential

**P/N** part number

**P/P** peak-to-peak voltage

**ppm** parts per million

**psig** pounds per square inch, gauge

**RAM** random access memory

**RF** radio frequency

**RMS** root mean square

**ROM** read-only memory

**RS-232** industry standard for serial communications

**s** second

**SIM** selected ion monitoring

**SRM** selected reaction monitoring

**TCP/IP** transmission control protocol / Internet protocol

**TIC** total ion current

**Torr** torr

**u** atomic mass unit

**UPS** uninterruptible power supply

**V** volt

**V ac** volts alternating current

**V dc** volts direct current

**vol** volume

**w** width

**W** watt

**WEEE** waste electrical and electronic equipment

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